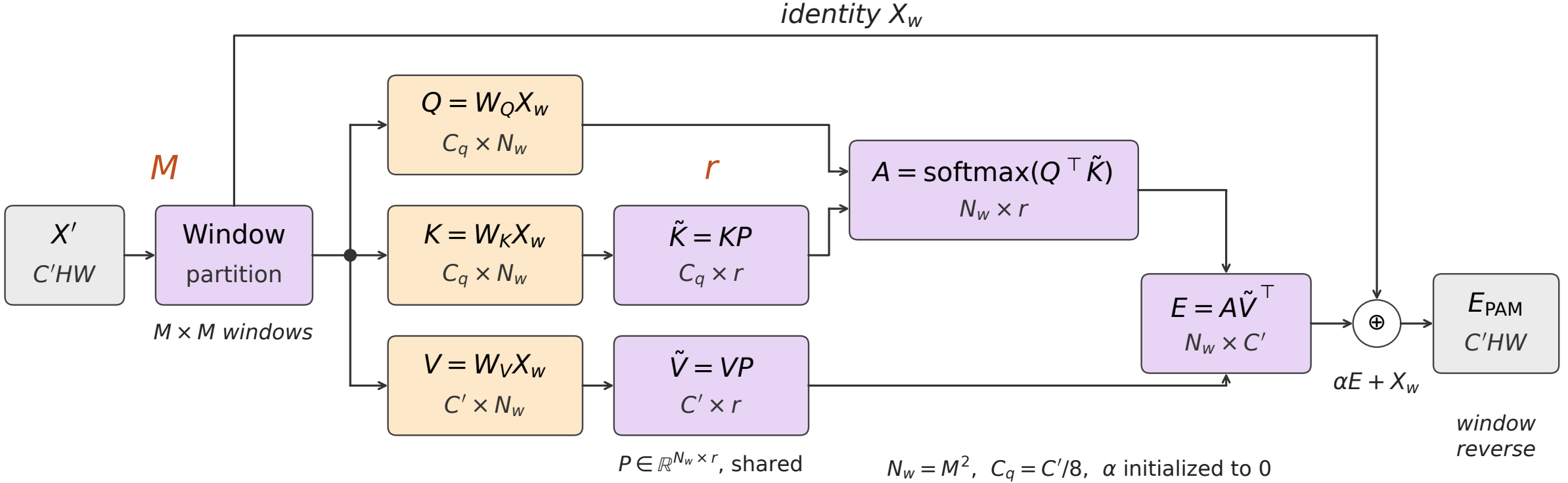
identity X_w
M
r
X'
$C'HW$
Window
partition
$M \times M$ windows
$Q = W_Q X_w$
$C_q \times N_w$
$K = W_K X_w$
$C_q \times N_w$
$V = W_V X_w$
$C' \times N_w$
$\tilde{K} = KP$
$C_q \times r$
$\tilde{V} = VP$
$C' \times r$
$P \in \mathbb{R}^{N_w \times r}$, shared
$A = \mathrm{softmax}(Q^\top \tilde{K})$
$N_w \times r$
$E = A\tilde{V}^\top$
$N_w \times C'$
$\oplus$
$\alpha E + X_w$
E_{PAM}
$C'HW$
window reverse
$N_w = M^2$, $C_q = C'/8$, α initialized to 0